Review: Solving one-step inequalities

1 Determine whether the following values satisfy the inequality $x+3>5$.

a $x=1$ b $x=3$ c $x=-9$ d $x=2$

2 Solve the following inequalities.

a $x+5\geq 11$ b $9>y+4$ c $n+8\leq 7$ d $m-5>9$

e $7\leq x-3$ f $x-8<11$ g $x-8\leq -4$ h $9y>81$

i $10x<90$ j $56\leq 8p$ k $3x\leq -18$ l $\frac{x}{3}\geq 6$

m $\frac{n}{7}>8$ n $8\geq \frac{x}{3}$ o $-4x<32$ p $-4y\geq -16$

q $-5x\leq -40$ r $-24<-8m$ s $\frac{a}{6}<11$ t $\frac{x}{-9}\geq 5$

u $\frac{x}{-7}<2$

3 For each of the following:

i Write down the inequality described by the statement.

ii Solve for the value of x.

a "Seven more than the value of x is at least nine".

b "Half of x is no more than four".

c "Eight is greater than the result of taking seven away from x".

d "Negative eight groups of x is less than three".

4 For each of the following:

i Solve the inequality.

ii Plot the solution to the inequality on a number line.

a $8+x\leq 8$ b $x+1<4$

c $x-4<1$ d $3+x<2$

e $-8<x-4$ f $-2+x\leq 4$

g $2+x\geq 7$ h $8<10+x$

i $-5+x<-7$ j $x-2\geq -3$

k $-4<-3+x$ l $6x\leq 48$

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m $36 \leq 9x$

$2x > -4$

o $7x < -35$

p $-5x \leq 15$

q $-3x > -21$

r $72 > -8x$

s $\frac{x}{2} \geq 7$

t $\frac{x}{6} < -2$

u $-6 \geq \frac{x}{3}$

v $\frac{x}{-8} \leq -2$

w $\frac{x}{-5} > 2$

x $\frac{x}{-7} < 2$

y $-2 \leq \frac{x}{-7}$

z $\frac{x}{3} > \frac{4}{3}$

5 For each of the following:

i Solve the inequality.

ii Plot the solution to the inequality on a number line.

a $x + \frac{7}{4} \geq \frac{9}{4}$

b $-4.5x \geq -18$

c $x + 2.5 \leq 7.5$

d $x + 3.5 > 8$

e $x - 2.6 \leq 2.4$

f $-4.5 + x \geq 4$

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